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Summary

My current research includes finding an efficient design for **Systems for ML (Especially LLMs)**, as well as employing **ML (And LLMs) models to optimize Systems**. Besides, I would like to conduct research on some hot areas like **Adapting distributed applications to an environment containing persistent memories**, **GPU scheduling of distributed applications**, in addition to **Software Hardware co-design** to optimize serverless computing environments.

Work Experience

August, 2020 – present **Research Assistant at Distributed System and Storage Lab, Virginia Tech.**

Advisor : Dr. Ali Butt, *Professor, Department of Computer Science, Virginia Tech*

- **Metis:** The project introduces a data access pattern for Deep Learning workloads to improve the cachability of the deep learning workloads. With a small cache, the project improves the cache hit ratio by up to 4.5× compared to the LRU caching policy.

- **User I/O profiling:** Led a collaborative research with Analytics & AI Methods at Scale Group at Oak Ridge National Laboratory (ORNL) on analytically recognizing the behavior of the users and jobs submitted to HPC systems to improve the I/O efficiency of the HPC systems. The analysis revealed that the I/O pattern of a users' job submissions **can be predicted over a window of 10 days with an accuracy of above 90%** based on the application-level I/O statistics and the applications' I/O patterns

- **LLMStore:** A collaborative research project with Analytics & AI Methods at Scale Group at Oak Ridge National Laboratory (ORNL) and Lawrence Berkeley National Laboratory **for LLM fine-tuning aiming to minimize offloading parameters, optimizer state and activation offloading in HPC using caching and deduplication solutions**. This project is carried out in collaboration with Jean Luca Bez, Ahmad Maroof Karimi, Arnab Kumar Paul, Suren Byna. Our initial analysis revealed **a traffic of over 300 GB for tensor offloading** to the storage for Llama finetuning.

June, 2024 – August, 2024 **Student Assistant at NERSC, Lawrence Berkeley National Laboratory (LBNL), internship.**

Mentors: Stephen Simms, Lisa Gerhardt, Jean Luca Bez

- I investigated the causes of I/O hotspots in HPC applications and analyzed common performance issues. Specifically, I examined Drishti, an HPC I/O recommendation tool, and found it generates many false positive warnings, and derived insights to improve the accuracy of the the Drishti I/O recommendation tool.

June, 2023 – August, 2023 **Student Assistant at Lawrence Berkeley National Laboratory (LBNL), internship.**

Mentors: Suren Byna, Jean Luca Bez

- I continued my research on characterizing the sources of I/O performance variation in HPC, and striving to alleviate the I/O performance variability. I found out a significant I/O variability for various HPC applications like E3SM (For earth modeling) and LAMMPP (For molecular simulations). Later continued my research for Large AI models and language models which I'm working on in the LLMStore project

- Presented a poster outlining my findings on the potentials for introducing I/O interference as one cause of variability
- Continuing my efforts to mitigate I/O interference in HPC systems for LLM fine-tuning/inference.

June, 2021 – **Internship at Oak Ridge National Laboratory, Analytics & AI Methods at Scale Group.**
August, 2021

Mentors: Feiyi Wang, Sarp Oral, Ahmad Maroof Karimi and Arnab Kamur Paul

- I studied and characterized the application I/O pattern using clustering techniques, and then extracted features from their submitter like the job runs of the same application over a time-window for that user, the scale of the job submissions the submitter tends to submit. **This work resulted in my ICDCN'25 paper, where I was able to predict the I/O pattern of the next job given the features from the past submissions of the same user with an accuracy of nearly 90% for HPC jobs.**

Notable projects

Fall 2023 – **LLMStore: An Efficient Activation/Parameter Storage Offloading Framework for LLM fine-tuning.**

- This project seeks to propose a **caching solution** for **LLM parameters/gradients/optimizer states** to **avoid read/writing (offloading) states and checkpoints into the slower tiers** like secondary storage. LLMs require a huge amount of memory across tiers like File system (Lustre in this case) and RAM to store the activations generated during the encode and decode phases along with my collaborators from ORNL and LBNL. Our initial observations suggest over 300 GB of traffic to Lustre for state offloading for LLama 70B, which further motivates the need to address the memory contention resulting from this huge traffic.

Fall 2020 – **SHADE: Enable Fundamental Cacheability for Distributed Deep Learning Training.**
Fall 2022

- A new **DLT-aware caching system** that detects **fine-grained importance variations at per-sample level** and leverages the variance to **make informed caching decisions** for a distributed DLT job. Mainly used Chameleon Cloud for the experiments. This results in a 4x hit rate improvement using a small cache compared to traditional algorithms like LRU.

Fall 2020 **Container layer caching evaluation for Docker.**

- Implemented and evaluated various eviction policies like LRU and LFU for Docker container layer caching as a course project for CS5204.

Computer skills

Programming Languages & frameworks	Python, R, C, C++, Advanced JAVA EE, JAVA SE, Tensorflow, Go, Rust, MPI, pthread, OpenMP, CUDA, Java bytecode, X86 assembly
Machine Learning & GPU programming	PyTorch, keras, scikit-learn, DeepSpeed, LLM, Computer Vision
Systems	Linux kernel programming, distributed systems, File Systems, Lustre, GPFS, Slurm, MapReduce, Redis, Ceph, NVMe, Darshan, Docker, Kubernetes

Conference & Workshop publications

[ICDCN'25] **Yazdani, Ahmad Hossein, Arnab K. Paul, Ahmad Maroof Karimi, Feiyi Wang, and Ali Butt.** User-based i/o profiling for leadership scale hpc workloads. In *Proceedings of the 26th International Conference on Distributed Computing and Networking, ICDCN '25*, page 181–190, New York,

NY, USA, Jan. 2025. Association for Computing Machinery. doi:10.1145/3700838.3700865.

[FAST'23★] Redwan Ibne Seraj Khan, **Yazdani, Ahmad Hossein**, Yuqi Fu, Arnab K Paul, Bo Ji, Xun Jian, Yue Cheng, and Ali R Butt. Shade: Enable fundamental cacheability for distributed deep learning training. In *Proceedings of the 21th USENIX Conference on File and Storage Technologies*, page 14, Santa Clara, CA, US, Feb. 2023. USENIX Association. URL <https://www.usenix.org/conference/fast23/presentation/khan>.

★ Top-tier venue

Education

- 2020–present **PhD, Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt
- May 2025 **Masters of Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt
- 2015–2020 : **Bachelor of Computer Software Engineering**, *University of Tehran*, Tehran, Iran.